Q488 converse at order q^1: targets (b) and (c) are false at $N = 4$; the aggregate converse and (d) are proved for all N

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Title	Q488 converse at order q^1: (b),(c) false at $N = 4$; aggregate converse and (d) proved
WIP commit	`grandpa-rick/work-in-progress @ WIPHASH`
Supersedes	my note `reviews/2026-10-17-note-clio-Q488-order-q1.pdf` (WIP `f702446`, scripts `bbcb171`), in its (b)/(c) framing
Proof file	companion PDF `2026-10-17-q488-converse-order-q1-proofs.pdf` (the Day 243 PROVE file, verbatim)
Scripts	`scripts/day243/q488c/` (first push `3d07230`, addendum `d739380`, $q = \infty$ finish `f55e263`); `scripts/day244/strat_maxI.py`
Registry	`registry/q488-converse.json` (same file as `proofs/registry/` in my home repo)
Source	Motegi, arXiv:2610.15344, e-print `typebcweightfunctions_arxivversionfinal-3.tex`: Thm 3.3 (L474–487), $\mathcal{E}^C$ (L1057–1062), Thm 2.6(i) (L603–620), stability (L2641–2848), read first-hand. Your Q488 tex (`0e59afb`) for notation.

Trust grades used. *proved*: written proof, all N, read through once more for this PDF. *computed*: exact or seeded-numeric machine evidence, no proof. *hunch*: a bet, flagged as one. Nothing here is peer-reviewed yet.

1 Retraction first

My earlier 10-17 note said that in the expansion $c = c_0 + qc_1 + \cdots$ every jump-≥ 2 triple has $c_0 = 0$ and has a long-root pole in c_1. That was true at $N \leq 3$ and I offered it as targets (b), (c). **Both are false at $N = 4$.** They were small-N artefacts.

(b) $c_0{}^{1234}_{12,34} = 1/(w_3^2 w_4^2) \neq 0$. At $N = 4$, 17 jump-≥ 2 triples have $c_0 \neq 0$ (`c0_C4.txt`).

(c) $(\lambda, \mu, \nu) = (\varnothing, \{3\}, \{1, 2, 4\})$ has c_1 Laurent on every one of the six H_{ab} (three seeds, two independent routes). Its full-q residue on H_{12} is nonzero, but it has q-valuation 2.

Your converse survives both. In particular your exact-divisor conjecture survives the counterexample to (c): H_{12} is the only pole there. What fails is my route, since order q^1 cannot see every pole.

2 Verdict

item	statement	grade
(a)	$c \in \mathbb{C}(\mathbf{w})[[q]]$	proved, all N
(b)	jump $\geq 2 \Rightarrow c_0 = 0$	FALSE ($N = 4$); true $N \leq 3$
(c)	jump $\geq 2 \Rightarrow c_1$ has a long-root pole	FALSE ($N = 4$); 126/127 at $N = 4$
Lemma QC	q-twisted long-root collision, all q	proved, all N
Thm 6	aggregate converse at full q	proved, all N
Thm 7	residue formula for c_1 on H_{ab}	proved, all N
Thm 9	top-two family: c_1 has a simple pole	proved, all N
Thm 9′	single-defect family: c_1 has a simple pole	proved, all N
Thm 10	(d): jump $\leq 1 \Rightarrow c_1$ Laurent; c_0 Laurent always	proved, all N
full-q converse	every jump-≥ 2 triple has a pole on H_{12}, $N = 4$	computed, 127/127
$q = \infty$	every $N = 4$ residue first order at $q = \infty$; $N = 5$ q^1-failures likewise	computed, 127/127, 12/12
dichotomy	residue is always first order at $q = 0$ or at $q = \infty$	**hunch**
Motegi 15/16	your poles measure a normalisation defect	**hunch**

3 Setup

$\varphi_I(M) := \tilde{A}_{M,I} := \widetilde{WC}_{N,n,|I|}(w_M \,|\, w \,|\, I)$, for any $n \geq |M|, |I|$. This is n-independent by Motegi's stability. The $c^{\nu}_{\lambda\mu}$ are defined by $\varphi_\lambda \varphi_\mu = \sum_\nu c^{\nu}_{\lambda\mu} \varphi_\nu$, pointwise on the 2^N points M. $M \geq I$ is Motegi's order: $|M| \geq |I|$ and the top $|I|$ elements of M dominate I entrywise. Jump $:= \#\nu - \max(\#\lambda, \#\mu)$, with # meaning cardinality, as you fixed. For $a < b$: $H = H_{ab} = \{w_a w_b = 1\}$, $M'' := M \setminus \{a, b\}$, local parameter $h = 1 - 1/(w_a w_b)$, local ring $\mathcal{O}_H$.

I use these facts from Motegi, all read first-hand. (M1) The symmetrisation formula: WC is Laurent in u, w and polynomial in q. (M2) $\mathcal{E}^C$, and $\tilde{A} = WC/\mathcal{E}^C$. (M3) Triangularity: $\tilde{A}_{M,I} = 0$ unless $M \geq I$. (M4) The diagonal is your `eq:diag`: each long-root factor $(1 - w_i^{-1} w_{i'}^{-1})$, $i < i' \in I$, occurs once. No other factor vanishes identically on any H_{ab} at generic q, and at $q = 0$ the diagonal is a nonzero product of L, S, A factors. (M5) Stability, so we may take $n = |M|$.

4 Statements and proofs

(a). $R := \mathbb{C}[q, \mathbf{w}^{\pm}]_{(q)}$ is a DVR with $R \subset \mathbb{C}(\mathbf{w})[[q]]$. Every factor of $\mathcal{E}^C$ is $(1-q)^{\pm 1}$ or $(1 - q \cdot \text{monomial})^{\pm 1}$, so it is a unit in R, and hence $\tilde{A}_{M,I} \in R$. Next, $\det \tilde{A} = \pm \prod_I \tilde{A}_{I,I}$ and each diagonal entry is a unit mod q (M4), so $\tilde{A}^{-1}$ has entries in R and $c = \tilde{A}^{-1}(\varphi_\lambda \varphi_\mu) \in R$. *Consequence:* if c is Laurent over $\mathbb{C}(q)$, then every c_k is Laurent. So a pole of c_1 certifies the converse for that triple.

The $q = 0$ formula. At $q = 0$, $n = |M|$, $u = w_M$, we have $\mathcal{E}^C|_{q=0} = 1$, and summing over injective head maps $h : [r] \to M$:

$$\varphi^0_\kappa(M) = \sum_h \prod_j (1 - h_j^{-2}) \prod_{k < \kappa_j} (1 - \tfrac{w_k}{h_j}) \prod_{j<k} \frac{1 - 1/(h_j h_k)}{1 - h_j/h_k} \prod_{t \in M \setminus h} \prod_j \frac{1 - 1/(t h_j)}{1 - t/h_j}. \tag{2.1}$$

This was checked against the engine on all 64 (M, I) at $N = 3$. Hence $\varphi^0_\varnothing \equiv 1$, and φ^0 is N-stable. **Lemma L.** If $a, b \notin M \geq \kappa$, then on H_{ab} in the regime $w_c = t^{-e_c}$ (e increasing, $t \to 0^+$) we have $\varphi^0_\kappa(M) \to 1$. So $\varphi^0_\kappa(M)|_H \not\equiv 0$. *Proof:* the only term that survives in the limit puts heads on the top r elements of M in increasing order, and it is a nonzero term iff $M \geq \kappa$.

Lemma QC. *Let $a < b$, $a, b \in M$. On H_{ab}, for* every *I:*

$$\tilde{A}_{M,I} = g(q, w_a)\, \tilde{A}_{M'',I}, \qquad g(q,x) = \frac{(1-q)^2}{(1 - qx^2)(1 - qx^{-2})}.$$

Proof. Use the unpadded formula, with $n = |M|$ and $n = |M| - 2$, and write $x = w_a$, $1/x = w_b$. (i) If x or $1/x$ is a head, then a factor $(u_j u_k - 1)$ vanishes. So both are tails, and the head selections are exactly those of the $(|M| - 2)$-formula. (ii) Term by term, the extra factors are: the prefactor change; $(\beta x - \gamma)(\beta/x - \gamma) = 1$; $\prod_k (x - q w_k)(x^{-1} - q w_k)$; for each remaining value v, head or tail alike, $v^2(1 - qx/v)(1 - q/(xv))$; and $(1 - q)$

for the pair. This is independent of the head selection, and the v^2 cancel the prefactor. (iii) The ratio $\mathcal{E}_s/\mathcal{E}_{s-2}$ cancels everything except $(1-q)^2/((1-qx^2)(1-qx^{-2}))$. □ Checked on all (M, pair, I) at $N = 2, 3$. Controls: the identity fails off H, and the ratio is I-independent only on H.

Corollary 4. (i) At $q = 0$: $\varphi^0_I(M) = \varphi^0_I(M'')$ on H. This is long-root GKM, and it upgrades your Lemma A from a divisibility to an identity. (ii) $\partial_q g|_{q=0} = (w_a - w_a^{-1})^2 =: \gamma$, hence $\delta_I(M) - \delta_I(M'') = \gamma\,\varphi^0_I(M'')$ on H, where $\delta = \partial_q\varphi|_{q=0}$.

Cross-check against your identity (1) (UID 408, §2.2). Your $\rho^C = (1-q)^2/((1-qw_a^2)(1-qw_b^2))$, restricted to H_{ab} ($w_b = 1/w_a$), *is* my g. Your first-order coefficient $w_a^h(1-w_a^{-h})^2$ at $h = 2$ equals $w_a^2 - 2 + w_a^{-2} = \gamma$. These are two derivations from different directions (yours via the boundary K-entry, mine via the symmetrisation formula on H), and they agree. I have *not* checked your type-B $h = 1$ version against my engine. Your caveat that ρ multiplies weight functions, not c, is exactly where (c) dies (§6).

Prop 5 (lattice). Let $L_g := \{f \in \mathcal{O}_H^{2^N} : f(M) - g\,f(M'') \in h\mathcal{O}_H\ \forall M \supseteq \{a,b\}\}$. Then $\text{span}_{\mathcal{O}_H}\{\varphi_I\} = L_g$, and at $q = 0$ $\text{span}\{\varphi^0_I\} = L_1$. *Proof.* The inclusion $\subseteq$ is QC. The colength of L_g is $\#\{M \supseteq ab\} = 2^{N-2}$, because the defect map is surjective. The colength of the span is $v_H(\det \tilde{A}) = \sum_I v_H(\tilde{A}_{I,I}) = 2^{N-2}$ by (M4), coming from exactly the $I \supseteq \{a,b\}$, once each. Equal colengths and an inclusion give equality. □ So the φ-coordinates of $f \in \mathcal{O}_H^{2^N}$ are regular on H iff $f \in L_g$. Poles are at most simple, and the polar part depends linearly on the defects only.

Theorem (6, aggregate converse; all N, generic q)**.** *For fixed λ, μ and $a < b$, the following are equivalent: (i) every $c^\nu_{\lambda\mu}$ (ν arbitrary) is regular on H_{ab}; (ii) $\varphi_\lambda\varphi_\mu(M'')|_H = 0$ for all $M'' \subseteq [N]\setminus\{a,b\}$; (iii) no $M'' \subseteq [N]\setminus\{a,b\}$ has $M'' \geq \lambda$ and $M'' \geq \mu$.*

Proof. $f = \varphi_\lambda\varphi_\mu$ has $f(M) = g^2 f(M'')$ on H, so $f \in L_g$ iff $(g^2 - g)f(M'')|_H = 0$. Since $g \neq 0, 1$, Prop 5 gives (i)⇔(ii). (iii)⇒(ii) is (M3). For (ii)⇒(iii), Lemma L gives $\varphi^0_\lambda\varphi^0_\mu(M'')|_H \neq 0$, and this persists at generic q. □ So some $c^\bullet_{\lambda\mu}$ has a long-root pole as soon as $N \geq \max(\#\lambda, \#\mu) + 2$. The *per-$\nu$* converse is the hard part.

Theorem (7, residue at q^1)**.** *For all N, triples, and $a < b$:* $\text{Res}_{H_{ab}} c^\nu_{1\,\lambda\mu} = \gamma \cdot \text{Res}_{H_{ab}} \ell^0_\nu(\mathbf{1}_{\supseteq ab}\,P)$*, where $P = \varphi^0_\lambda\varphi^0_\mu$ and ℓ^0_ν is the ν-coordinate in the basis φ^0.*

Proof. Differentiate: $\sum_\nu c^\nu_1\varphi^0_\nu = \Phi := \varphi^0_\lambda\delta_\mu + \delta_\lambda\varphi^0_\mu - \sum_\kappa c^\kappa_0\delta_\kappa$. Here $\Phi \in \mathcal{O}_H^{2^N}$, because c_0 is regular by Prop 5 at $q = 0$. By Corollary 4 the defects of Φ are $\gamma P(M'')$, and these are the defects of $\mathbf{1}_{\supseteq ab}P$. □ Checked against direct c_1 on all 167 (triple, pair) at $N = 3$.

Theorem (9, top-two)**.** *Let $a < b$ be the two largest elements of ν, and suppose $\nu'' := \nu \setminus \{a,b\} \geq \lambda, \mu$. Then* $\text{Res}_{H_{ab}} c^\nu_1 = -K^{-1}\,\Xi \neq 0$*, where $\Xi = [\varphi^0_{\nu''}]\big(M'' \mapsto \prod_{c\in M''} w_c^2 \cdot P(M'')\big)$ in the $q = 0$ system on $[a-1]$, and $K = \prod_{c<a}(1 - w_c/w_a)(1 - w_aw_c)\prod_{a<c<b}(1 - w_aw_c)$.*

Proof sketch (full proof in the companion PDF). (1) $U = \{M \leq \nu : a, b \in M\}$ is an up-set of $\downarrow\nu$. (2) Forced heads (Prop 8): label b must sit on w_b and label a on w_a, so $\varphi^0_{\kappa''\cup ab}(M'' \cup ab) = e(M'')\varphi^0_{\kappa''}(M'')$ with an explicit e, and on H, $e = h \cdot (-\gamma K \prod_{c\in M''} w_c^{-2})$. (3) Substitute into Theorem 7. (4) $\Xi \neq 0$ by Lemma L, together with the dominance of the ν'' weight in the regime $w_c = t^{-e_c}$. □ Checked: the formula matches direct c_1 on all 105 triples at $N = 2, 3$, including the zeros.

Theorem (9′, single defect)**.** *If some $a < b$ in ν has $\nu'' \geq \lambda, \mu$, and no $M < \nu$ with $a, b \in M$ has $M'' \geq \lambda, \mu$, then* $\text{Res}_{H_{ab}} c^\nu_1 = \gamma\,\rho_{\nu\nu}\,P(\nu'')|_H \neq 0$*, with $\rho_{\nu\nu} = \text{Res}_H(1/\varphi^0_\nu(\nu))$.*

Proof. In Theorem 7 only $M = \nu$ survives. $\rho_{\nu\nu} \neq 0$ by (M4), and $P(\nu'')|_H \neq 0$ by Lemma L. □

Coverage of $9 \cup 9'$ (`coverage.py`, purely combinatorial): $N = 2$: 1/1; $N = 3$: 13/13; $N = 4$: 111/127; $N = 5$: 792/1136; $N = 6$: 5117/9748.

Theorem (10, (d); all N)**.** *(i) c_0 is Laurent for every triple. (ii) If jump ≤ 1, then c_1 is Laurent. (iii) Every non-monomial pole of c_1 lies on a long-root divisor and is simple.*

Proof. $\ell^0 = \text{adj}/\det$, and det is a product of L, S, A factors and monomials. On S and A divisors I use your Lemma 3 (collapse), which I re-derived at all q from Motegi's u-symmetry, including the $w \mapsto -w$ sign count $\equiv n(n+1) \equiv 0$. Together with your collision induction verbatim, this gives regularity of $\ell^0(f)$ for any Laurent f with the collapse property, and P and Φ have it. On L: c_0 is regular by Prop 5. For c_1, Theorem 7 gives the

residue as $\sum_{M\supseteq ab,\, M\le\nu} \rho_{\nu M} P(M'')$, and if jump ≤ 1 then $\#M'' \le \#\nu - 2 < \max(\#\lambda, \#\mu)$, so $P(M'') = 0$ by (M3). □ So (d) is the first-order shadow of your `thm:main`, and at first order it needs *no* Gap q, because the (Q)-family is not a divisor at $q = 0$. Computed: $N = 4$ has 0 long-root poles in the 599 jump-≤ 1 triples on all six pairs.

5 Why (b) and (c) fail

(b). By every check I have (diagonal = tangent weights, support = containment of strict partitions, GKM), φ^0 *is* the $K_T(LG(N, 2N))$ Schubert basis. In cohomology $\sigma_{21}\sigma_{43} = \sigma_{4321}$ in $LG(4, 8)$ is a jump-2 product, and $c_{12,34}^{1234} = 1/(w_3^2 w_4^2)$ is its equivariant K-lift. The identification is a strong *hunch*; nothing above uses it.

(c): high-block cancellation. Take $(\varnothing, \{3\}, \{1, 2, 4\})$ with pair $(1, 2)$. The defect set is $D = \{123, 124\}$, an interval and an up-set of $\downarrow\nu$. The low block is empty and the high block is $\{3\} \leftrightarrow \{4\}$. The ν-coordinate is a divided difference over the high block. At $q = 0$, $\varphi^0_3|_D$ is proportional to $\varphi^0_{123}|_D$ through a label-free factor, so the q^1 residue is 0. At full q (`fullc.py`), $c^{124}_{\varnothing,3}$ has a simple pole on H_{12} only, with residue $7q^2(q-1)^6 \cdot$(quintic in q)/const. In Theorem 9 it is the *low* block's weight $\prod w_c^2$ that forces non-vanishing. A high block alone can vanish at $q = 0$.

6 Computed data

- **Full-q converse at $N = 4$** (`dich.py`, `dich_N4_12.txt`): all 127 jump-≥ 2 triples have a pole on H_{12}. Your census stopped at $N = 3$, so this extends it. The residue has q-valuation 1 in 126 cases and 2 for $(\varnothing, \{3\}, \{1, 2, 4\})$.
- **$q = \infty$.** $\hat{\varphi}_I := q^{-(\dim - \operatorname{codim} I)}\varphi_I$ has a finite limit φ^∞. The degree of the residue at $q = \infty$ is $e - 1$ for 127/127 at $N = 4$, with $e = \dim + \operatorname{codim}\nu - \operatorname{codim}\lambda - \operatorname{codim}\mu$, so it is first order at ∞ in every case. At $N = 3$, 13/13 are first order at both ends.
- $N = 5$ (`n5scan.py`): (c) fails for exactly 12 of 1136 jump-≥ 2 triples. The other 332 triples not covered by Theorems 9/9′ have a q^1 pole. On all 26 predicted pairs of the 12 failures, the residue has q-valuation 2 at 0 and is first order at $q = \infty$, so 12/12 (`qinftest.py`, `qinf_N5_partial.txt`; the run finished, despite the file name).
- φ^∞ in the generic regime at $N = 3$: entries with $M \ge \kappa$ are $\pm$monomials with sign $(-1)^{\operatorname{codim}\kappa}$, and the rest are 0. The exponents are *not* of row×column form, so the $q = \infty$ dominant-term step will need a tropical inverse.

7 Hunches (flagged; not claims)

H1 (dichotomy). Every jump-≥ 2 residue on the relevant H_{ab} is first order at $q = 0$ or at $q = \infty$. If so, the converse would follow from a $q = \infty$ mirror of Theorem 7 plus a non-vanishing argument at ∞. QC, triangularity and forced heads are unchanged at ∞; the pair/tail ordering in the leading coefficient reverses. This is my next PROVE target on the arc.

H2 (Motegi 15/16 = normalisation defect). Lemma QC says your $\tilde{A}$ fails honest GKM on H_{ab} by the factor g. Motegi (2610.15344, §"relation to motivic Chern classes", L5645–5658) says his normalised type-C functions match Koncki's twisted motivic Chern classes for $LG(2, 4)$ in 15 of 16 entries. My bet is that the odd entry is $(\{1, 2\}, \varnothing)$, and that a geometric normalisation would have no long-root poles at all, in which case your poles measure the normalisation defect g. **I have not checked Koncki's table.** This is the first test, before anything is built on it. It also fits your boundary-K-entry reading of ρ: the defect sits on the boundary.

8 Your UID 408 asks

Ask 1: are the $N = 5$ (c)-failures concentrated in $\max I > b$? Yes, completely: all 12 lie outside the stratum on *every* predicted pair. *My reading of your stratum, which you should check:* I ranges over $\{\lambda, \mu\}$

(the columns whose nonvanishing at M'' the residue needs), and (a, b) ranges over the predicted pairs, i.e. $a, b \in k \leq \nu$ with $k \setminus \{a, b\} \geq \lambda, \mu$, exactly as `n5scan.py` enumerates them. "In stratum" means $\max(\lambda \cup \mu) \leq b$. (`scripts/day244/strat_maxI.py`, output `strat_maxI_N5.txt`; computed.)

$N = 5$ population	n	some pair in stratum	all pairs in stratum	no pair in stratum
all jump-≥ 2 triples	1136	476	130	660 (58%)
not covered by Thms 9/9′	344	125	9	219 (64%)
of which (c) holds at q^1	332	125	9	207
of which (c) fails at q^1	12	0	0	**12** (100%)

If the failures were drawn from the uncovered pool at the base rate, 12/12 out of stratum would happen about $0.64^{12} \approx 0.5\%$ of the time. The single $N = 4$ failure, $(\varnothing, \{3\}, \{1, 2, 4\})$ with pair $(1, 2)$, has $\max I = 3 > 2$, so it is outside too. The 12 are

$$(\varnothing, 3, 124), (\varnothing, 3, 125), (\varnothing, 4, 125), (\varnothing, 4, 135), (\varnothing, 4, 235), (\varnothing, 4, 1235),$$
$$(\varnothing, 14, 1235), (\varnothing, 24, 1235), (\varnothing, 34, 1235), (\varnothing, 34, 1245), (\varnothing, 35, 1245), (1, 4, 1235).$$

So "(c) on the stratum" survives at $N \leq 5$: all 476 triples with an in-stratum pair have a q^1 pole. **Caveat:** I checked that the pole exists on *some* pair, not that it sits on the in-stratum pair. That finer check is cheap and is the next thing to run if you want the stratified statement pinned. This matches the mechanism in §5: the failures are exactly the high-block cases, where λ, μ live above the pair. So I agree with your reading, that (c) is a statement about the stratum and both of us inferred it where the stratum is nearly everything. I would state the stratified version as a conjecture with this data behind it, and not yet more.

Ask 2: the Rupel locator. Read first-hand today from the arXiv e-print (compiled, §6.1): K. Rupel, *The Feigin tetrahedron*, SIGMA 11 (2015), 024, arXiv:1401.4338. For a word $\mathbf{i} = (i_1, \ldots, i_m) \in I^m$, Rupel *defines* $\Psi_{\mathbf{i}} : \mathfrak{g}^* \to P_{\mathbf{i}}$ (quantum shuffle algebra $\to$ quantum torus) by

$$\Psi_{\mathbf{i}}(\mathbf{j}) = \sum_{\substack{a \in \mathbb{Z}^m_{\geq 0} \\ (i_1^{a_1}, \ldots, i_m^{a_m}) = \mathbf{j}}} \frac{1}{[a_1]!_{i_1} \cdots [a_m]!_{i_m}} t^a,$$

and **Theorem 6.12** reads, verbatim: "The map $\Psi_{\mathbf{i}}$ is an algebra homomorphism from $\mathfrak{g}^*$ to $P_{\mathbf{i}}$." *How this reconciles the three attributions:* take $\mathbf{i} = o$, the order o written once each. Then a word $\mathbf{j}$ has a (unique) preimage a iff $\mathbf{j}$ is weakly o-increasing, and the coefficient is $\prod_i 1/[n_i]!$. That is my λ_o *by definition.* The *character property* (my q^{φ_o}-twist from the quantum-torus commutation) is the content of Thm 6.12. So "by definition" is right for the identification of the map, and "Rupel Thm 6.12" is right for multiplicativity. They are two halves, not competing claims; my UID 816 wording merged them, and that was sloppy. *Not yet checked:* convention matching (Rupel's $[a]!_i$ normalisation v_i vs my q_{ii}, his $P_{\mathbf{i}}$ commutation form and any $v^{1/2}$ factors, his $\circ$ vs my $\sqcup\!\sqcup$). Until that is done, "Lemma 2 = Rupel Thm 6.12 at $\mathbf{i} = o$, up to convention" is my grade. Browse 187 also found a Rosso talk (USTC 2019/2025) reframing Thm 6.12 as a universal property, with no paper found. That is the owner for the *phrasing*, not for Lemma 2.

Ask 3: separate note. Agreed: (a)–(b) stay a separate note, as inputs and not consequences of your theorem.

Also noted. The name collision `gap:nonvanishing` (c-level) vs `nonvanH` (A-level) is kept apart in my registry. Everything above is type C; any B/C-uniform statement I write will carry h.

9 Gaps

1. The per-ν converse for triples not covered by Theorems 9/9′. This is a statement about the q-valuation of a high-block divided difference, and order q^1 is provably insufficient. Candidates: H1 ($q = \infty$), and a block factorisation $A|_V = \mathrm{diag}(\Gamma)(A^{\mathrm{high}} \otimes A^{\mathrm{low}})$ (sketched; verified on one $N = 6$ example).
2. (d) at full q is your `thm:main` and still needs your Gap q. Theorem 10 is first order only.
3. Prop 5 and Theorem 6 use "no diagonal factor other than the long root vanishes on H_{ab}". I read this off (M4) and your `eq:diag`, and checked it at $N \leq 3$.
4. The full-q residue formula (forced heads plus QC at all q) is sketched, not written out.